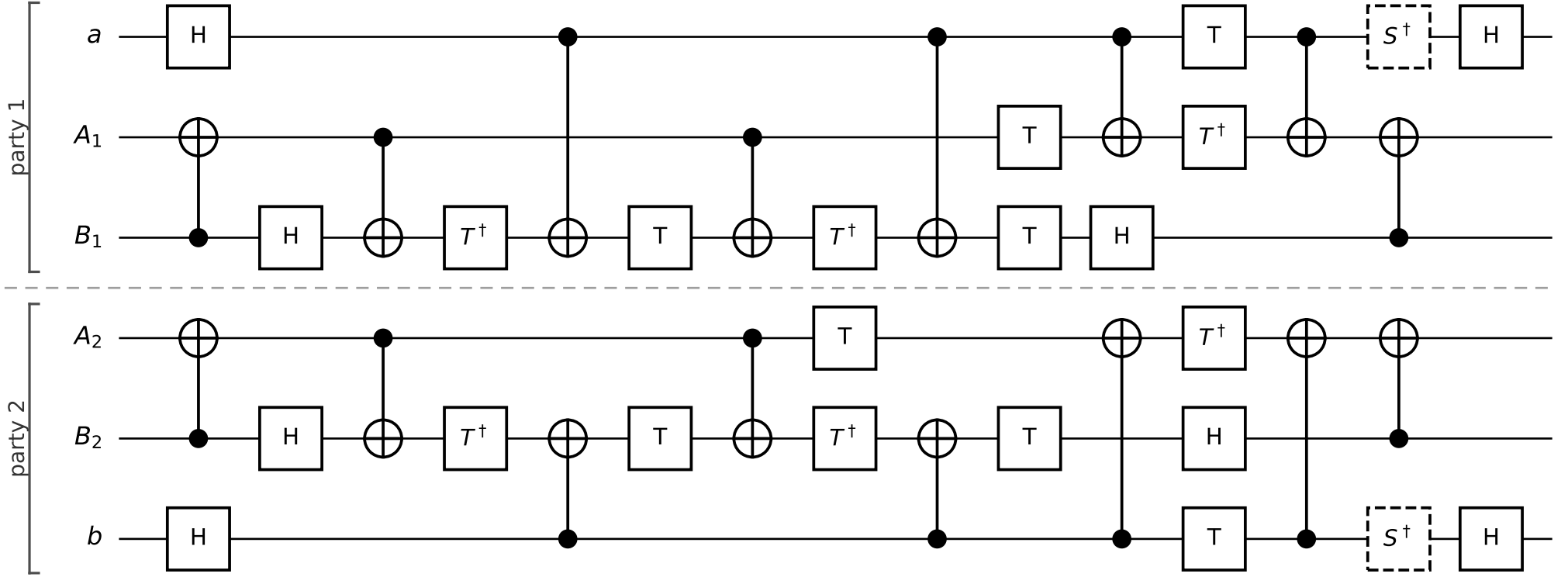
6Q-L16 (Type 3): two local textbook Controlled-SWAPs (16 CNOTs)



Each party runs the Step-3 textbook Clifford+T Controlled-SWAP (8 CNOTs) on its own ancilla;

no two-qubit gate crosses the party cut. A two-qubit replacement depolarizing channel D_q is applied after every CNOT.

Dashed $S^\dagger$: present only in the YY setting. The estimator $\Delta(O) = \langle X_a X_b \otimes O \rangle - \langle Y_a Y_b \otimes O \rangle$ requires both settings.